

Digital Nudging in Mobile App Onboarding: Field Evidence on User Engagement

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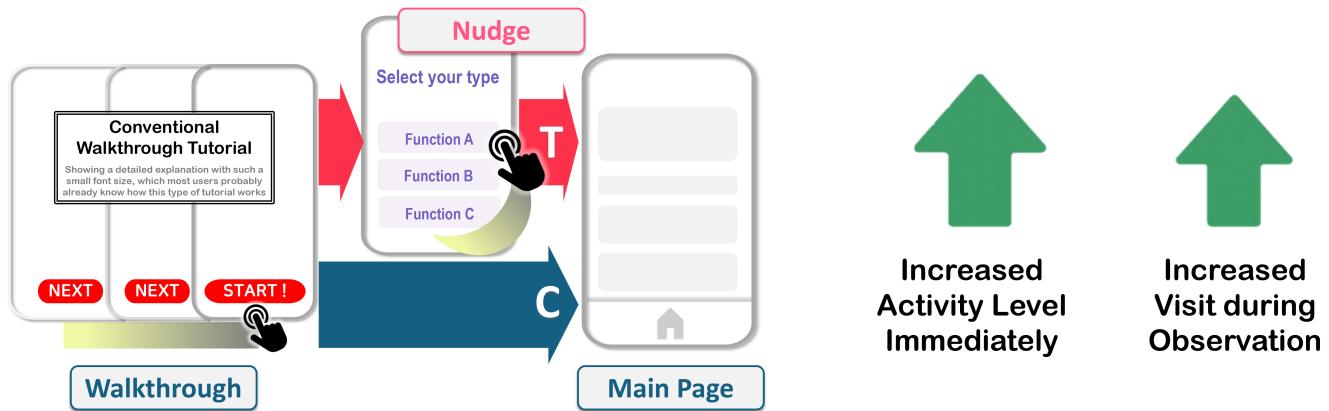


Figure 1: Overview of the experiment design and results. Users complete the walkthrough after app installation and registration. Then, treatment group users receive a prompt to select a usage type, while control group users move directly to the next step without this intervention. Next, users in both groups enter the main page and begin using the app. As results, treatment group users exhibited significantly higher activity levels and more frequent visits during the observation period. The intervention was effective in both the short and long term.

Abstract

Onboarding strategies often prioritize initial adoption in competitive mobile app ecosystems, but may undermine long-term engagement. This study examines whether a minimal intervention during onboarding can influence long-term behavior. We conducted a large-scale field experiment with thousands of users over several months, where the treatment group was asked to select a usage type on the final screen. Our results show that the treatment group exhibited significantly higher activity levels and more frequent visits immediately after sign-up and throughout the observation period. While the effect gradually diminished, it remained significant. The intervention likely activated System 2 thinking, helping users become more aware of app functions, while the doorway effect may have caused users to forget previous content during the onboarding transition. This study suggests that a low-cost, sense-of-agency-preserving intervention can drive sustained user engagement, contributing to the literature on digital nudging and user behavior. Our findings also provide empirical evidence for

future technical development and highlight the ethical potential of digital nudging in fostering user agency.

Keywords

User Engagement, Doorway Effect, Dual Processing Theory, Digital Nudge, Field Experiment

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1 Introduction

Smartphones have become the most widely used digital interface with approximately 4.88 billion users worldwide as of 2024. Global revenues from mobile apps has been growing and reached \$420.8 billion in 2022, and consequently firms spending over \$87.6 billion on advertisements alone to lead to installation and acquire users. In the meantime, developers have continually advanced features like autoplay, infinite scroll, and recommendation systems in efforts to attract and retain users, enhancing engagement and convenience.

However, indiscriminate investment and development often result in harm to both companies and users. To maximize opportunities and resources, it is crucial to explore effective ways to engage

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Table 1: Treatment Effects on Daily Activity

DV	Overall	Daily Activity		
	Activity	Day 1	Day 2	Day 14
Treat	***	***	***	***
Sign-up FE	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes

Note: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

users on small screens. Platforms frequently employ subtle behavioral design strategies—commonly referred to as digital nudges—to influence user choices and guide interaction flows. Thaler and Sunstein [8] define nudges as choice-preserving modifications of the decision environment that steer behavior in predictable ways. In digital contexts, these nudges include visual defaults, simplified interfaces, and guided interaction paths [7]. Despite their popularity, there are growing concerns about the ethical implications of such designs, particularly when they prioritize attention capture over user welfare [1, 2]. Overreliance on frictionless, attention-maximizing features can lead to passive consumption and eventual disengagement [4, 5].

In response, this study focuses on creating meaningful engagement from the beginning. Industry reports suggest that over 20% of users abandon an app after the first use, and fewer than 10% remain active after one month. This highlights a gap in many onboarding strategies, which prioritize short-term conversion over sustained value creation. We thought that behavioral designs that prompt user reflection and intentional usage, as opposed to passive exposure, could offer a more sustainable path to long-term engagement. Conventional walkthroughs, as explained by Jakob’s Law, often rely on repetitive, intuitive actions, leading users to engage with familiar buttons in a predictable manner.

This experiment is grounded in two key psychological mechanisms:

Dual Processing Theory. System 1 involves intuitive, effortless thinking, while System 2 requires thoughtful decision-making [3]. **Doorway Effect.** Transitions between screens can lead to forgetfulness. In our setting, we assumed that the transition between tutorial screens may reinforce forgetfulness due to this effect [6].

We hypothesized that, although our intervention might increase cognitive load for users, it likely activated System 2 thinking by prompting them to make a choice, thus making them more aware of the app, even when disrupted by the doorway effect during the walkthrough.

Although simple, this experiment involved thousands of users over several months, showing that even small changes in a familiar tutorial structure can have significant effects. While much research focuses on controlled environments, this study offers empirical evidence from a large-scale field setting, bridging theory and practice. It demonstrates that minimal, low-cost interventions during onboarding can significantly enhance user engagement without complex designs. Our findings challenge the idea that advanced technologies or frictionless designs are the key to user retention, suggesting instead that fostering reflection and early user agency leads to better long-term engagement.

2 Experiment

The experiment was conducted in a field setting during the onboarding process. Upon app installation and registration, users were randomly assigned to either a treatment group or a control group, with a 50:50 distribution. All users, regardless of group assignment, were exposed to a standard tutorial consisting of multiple pages that introduced the core features of the app. Users progressed through the tutorial by tapping buttons consistently located at the bottom of the screen. The intervention was introduced at the final stage of the tutorial. In the treatment group, users were asked to select one of the core functions, presented as large buttons in the center of the screen, to proceed to the main page. In contrast, users in the control group moved directly to the main page without making a choice. Our analysis includes at least a hundred thousand users who completed the sign-up process for the focal app during several months of experimental period, ensuring that each user has at least two weeks of post-sign-up activity available for analysis.

3 Result

In our OLS regressions, we found that the intervention significantly increased both total activity and number of visit. The treated users performed more actions and visited the app more frequently during the period, with these effects statistically significant and robust to potential imbalances and covariates, as well as sign-up week fixed effects. The intervention, prompting users to select a key function during onboarding, boosted user engagement in terms of both depth and duration.

The treatment had the strongest impact immediately after the intervention, on the first day after sign-up. Although the effect gradually decreased, it remained positive and statistically significant over time. All daily estimates were highly significant, demonstrating the robustness of the time-varying effect.

4 Conclusion

This study examines how a simple, single-page intervention affects user behavior both immediately and over time. Using a field experiment embedded in the onboarding process, we find that this prompt significantly increased activity levels and visit frequency. Additionally, we hope it provides an empirical foundation for future technical development.

Our findings contribute to ongoing discussions about digital nudging and its ethical implications. Unlike some intrusive nudges that may limit user agency, our intervention encouraged users to reflect on their intended use, fostering self-determined action and a sense of agency. This approach not only boosted short-term engagement but also enhance long-term user loyalty, by supporting users’ sense of control.

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